

AANAND MAHAWAL

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LinkedIn

Portfolio

GitHub

LeetCode

GFG

EDUCATION

National Institute of Technology, Kurukshetra

Aug 2022 – May 2026

B.Tech in Electronics and Communication Engineering | CGPA: 8.51/10

EXPERIENCE

Research Intern | NIT Kurukshetra

Jun 2025 – Jul 2025

- Implemented the Affine lightweight block cipher in Python to encrypt large binary inputs (100,000 bits) into secure ciphertext.
- Conducted NIST statistical tests to evaluate randomness, entropy, and distribution properties of the encrypted output.
- Authored modular and testable Python code with clear documentation, emphasizing correctness, performance, and reproducibility.

SKILLS

Languages: Python, SQL, Java, C++, JavaScript

Machine Learning & Deep Learning: Scikit-learn, XGBoost, TensorFlow, PyTorch, NLP, Feature Engineering, Predictive Modeling

Generative AI: LLMs, RAG, AI Agents, LangChain, LangGraph, Prompt Engineering, FAISS, Groq API

Data Science: Pandas, NumPy, EDA, Statistical Analysis, Hypothesis Testing

Deployment & Databases: FastAPI, REST APIs, Streamlit, Docker, MLOps, Git, MySQL, MongoDB

Web Development: HTML, CSS, JavaScript, React

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

PROJECTS

Multi-Agent AI Resume & Interview Copilot

Live | [GitHub](#)

- Orchestrated **5 specialized AI agents** (ATS scoring, resume analysis, job-description matching, interview-question generation, career advice) over a shared **LangGraph** state graph with **LangChain** and Groq-hosted LLM inference.
- Grounded agent outputs in a **FAISS**-backed **RAG** pipeline over parsed PDF resumes, producing structured JSON reports with ATS scores, skill-gap analysis, and recruiter-style feedback instead of free-form text.
- Shipped the workflow as a **Streamlit** web app with PDF ingestion and tailored interview questions for each target role.

WhatsApp Chat Analyzer with Tool-Calling AI Agent

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- Engineered a **tool-calling LLM agent** (Groq) that answers natural-language questions about uploaded chats by routing each query to one of **15 deterministic Pandas functions**, preventing hallucinated counts, rankings, and quotes.
- Added lightweight **TF-IDF retrieval** for topic and summary questions — a hybrid **RAG** design with no vector database — along with automatic date-format detection and multi-turn conversation memory for follow-up questions.
- Optimized prompts, tool schemas, and result sizes to fit **Groq free-tier token limits**, with graceful recovery from rate-limit and oversized-request API errors.

RiskCare — ICU Risk Prediction & Medical AI Assistant

Live | [GitHub](#)

- Trained an **XGBoost** classifier that estimates ICU admission risk from five patient vitals (heart rate, blood pressure, temperature, respiratory rate, comorbidity index) at **95% accuracy**, served with real-time inference and risk-factor explanations.
- Paired the predictor with a **Groq**-powered medical chatbot for symptom explanation and patient education, with safety guardrails and a theme-locked, accessibility-focused **Streamlit** interface.

End-to-End MLOps Customer Churn Platform

Live | [GitHub](#)

- Built a modular **MLOps** pipeline (preprocessing, training, evaluation, inference) around a **Random Forest** model on the IBM Telco dataset, reaching **78.6% accuracy** and **64.6% F1** on churn prediction.
- Exposed the model through a **FastAPI** REST service with a **Streamlit** retention-analytics dashboard, **Dockerized** both, and deployed to **Render** and Streamlit Cloud for live scoring.

ACHIEVEMENTS & LEADERSHIP

- Secured an on-campus Data Scientist offer from Tredence.
- Solved 1000+ DSA problems across LeetCode and GeeksforGeeks.
- Oracle Cloud Infrastructure 2025 Certified Data Science Professional and Oracle Cloud Infrastructure 2025 Certified Generative AI Professional.
- Led peer-learning initiatives under Shiksha and contributed to NSS community activities.